

數字經濟核心科技系列：前端網站開發人員課程

(一) 初階網絡程式設計

3. 疊層樣式表(CSS3)

8. CSS: 高階III

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8. CSS高階III

學習成果

- 在本課中，你將學習：
 - ✓ 計數器
 - ✓ 佈局
 - ✓ 屬性選擇器
 - ✓ 圖像設計

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8. CSS高階III

8.1 計數器

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8.1 計數器

- 一個網頁可能包含很多信息，我們可能需要將這些信息分成不同的點。
- 在HTML中，我們可以使用一個列表來實現這一目的。

```
<ol>  
  <li>Using CSS Counters:</li>  
  <li>HTML Tutorial</li>  
  <li>CSS Tutorial</li>  
  <li>JavaScript Tutorial</li>  
</ol>
```

1. Using CSS Counters:
2. HTML Tutorial
3. CSS Tutorial
4. JavaScript Tutorial

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8.1 計數器

- 然而，我們可以使用函式計數器。
- 它類似於JavaScript中的"print"函式
- 我們可以定義要顯示的文字

Using CSS Counters:

Section 1: HTML Tutorial

Section 2: CSS Tutorial

Section 3: JavaScript Tutorial

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8.1 計數器 範例

對於計數器，我們可以使用更簡單的編碼來獲得類似的列表結果

。

```
<style>
body {
  background:lightgreen;
  counter-reset: section;
}

h2::before {
  counter-increment: section;
  content: "Section " counter(section) ": "
}
</style>
```

Using CSS Counters:

Section 1: HTML Tutorial

Section 2: CSS Tutorial

Section 3: JavaScript Tutorial

```
<h1>Using CSS Counters:</h1>
<h2>HTML Tutorial</h2>
<h2>CSS Tutorial</h2>
<h2>JavaScript Tutorial</h2>
```

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8.1 計數器 範例

```
body {  
    background:lightgreen;  
    counter-reset: section;  
}
```

"counter-reset"是必要的，以便將計數器分配到不同的部分。

Using CSS Counters:

Section 1: HTML Tutorial

Section 2: CSS Tutorial

Section 3: JavaScript Tutorial

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8.1 計數器 範例

"::before"，是用來定義計數器的顯示方式的。
我們可以通過

::before

指定元素。

```
h2::before {  
  counter-increment: section;  
  content: "Section " counter(section) ": ";  
}
```

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8.1 計數器 範例

“Counter-increment”是定義要計數的變數。

```
h2::before {  
  counter-increment: section;  
  content: "Section " counter(section) ": ";  
}
```

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8.1計數器 (總結)

訂列表

- 重複的程式碼 (CSS/HTML)
- 定義每個不同項目的不同樣式可能很耗時

HTML :

值</ LI>

值</ LI>

計數器

- 簡單程式碼 (CSS)
- 適用於內容
- 只需要創建內容

- CSS: “::before{
 1. counter-increment: value;
 2. “display message”
counter(value); }”

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8.1 計數器添加 (HTML)

計數器可以配合列表

```
<ol>
  <li>section A </li>
  <li>section B

  <ol>
    <li>section B1</li>
    <li>section B2</li>
    <li>section B3
  </ol>
```

```
    <li>section B1a</li>
    <li>section B1b</li>
    <li>section B1c</li>
  </ol>
  </li>
  <li>section B4</li>
</ol>
<li>section C</li>
<li>section D</li>
</ol>
```

```
1 section A
2 section B
  2.1 section B1
  2.2 section B2
  2.3 section B3
    2.3.1 section B1a
    2.3.2 section B1b
    2.3.3 section B1c
  2.4 section B4
3 section C
4 section D
```

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8.1 計數器添加 (CSS)

```
<style>

  body{
    background: lightseagreen;
    font-size: 20px;
  }

  ol {
    counter-reset: section;
    list-style-type: none;
  }

  li::before {
    counter-increment: section;
    content: counters(section, ".") " ";
  }

</style>
```

1 section A
2 section B
 2.1 section B1
 2.2 section B2
 2.3 section B3
 2.3.1 section B1a
 2.3.2 section B1b
 2.3.3 section B1c
 2.4 section B4
3 section C
4 section D

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8.1 計數器 活動5

創建一個有兩個標題部分和三個子部分的内容列表，如下所示。

1. 第一部分
2. 第二部分
 - 2.1 節2.1
 - 2.2 第2.2節

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8.1 計數器 活動5_建議的答案

僅供參考:

```
<style>

  body{
    background: lightseagreen;
    font-size: 20px;
  }
  ol {
    counter-reset: section;
    list-style-type: none;
  }

  li::before {
    counter-increment: section;
    content: counters(section, ".") " ";
  }

</style>
```

1 The first section
2 The second section
 2.1 section 2.1
 2.2 section 2.2

```
<body>
  <ol>
    <li>The first section </li>
    <li>The second section</li>

    <ol>
      <li>section 2.1</li>
      <li>section 2.2</li>
    </ol>
  </ol>
</body>
```

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3. 疊層樣式表(CSS3)

8.2 佈局

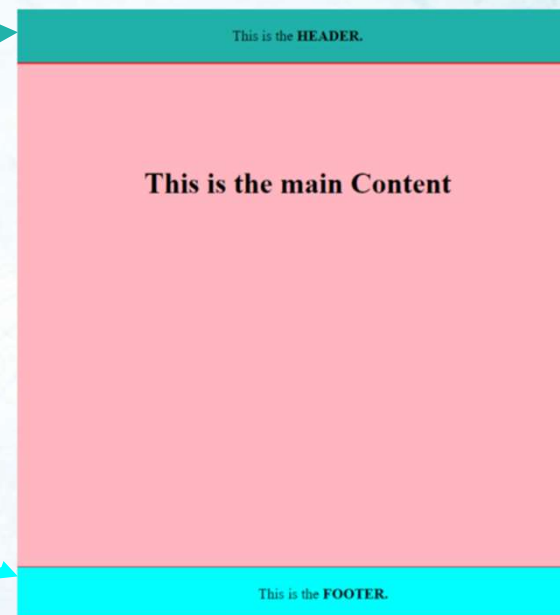
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8.2 佈局

一個網站可以被分為不同的部分

- 頁首
- 主要內容
- 頁腳



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8.2 佈局 (頁首) 示例

我們可以指定不同的 Header 樣式

```
}  
.header {  
  background-color: lightseagreen;  
  padding: 20px;  
  text-align: center;  
}
```

```
<body>  
  <div class="header">  
    This is the <strong>HEADER.</strong>  
  </div>  
</body>
```

This is the **HEADER.**

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8.2 版面設計(頁腳)示例

我們可以指定不同風格的頁腳

```
.footer {  
  width: 100%;  
  background-color: cyan;  
  position: fixed;  
  bottom: 0;  
  margin: 0px;  
  padding: 20px;  
  text-align: center;  
}
```



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8.2 佈局

- 對於頁首和頁腳，它類似於一個DIV。
- 我們可以使用同樣的屬性來設計一個合適的界面。
- 佈局對於一個網頁的設計很重要，因為網頁的笨拙設計可能會影響其可讀性。

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8.2佈局添加（列）

為了很好地展示信息，我們可以將內容分為不同的列

Link1 Link2 Link3		
column 1	column 2	column 3

For reference: https://www.w3schools.com/css/css_website_layout.asp

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8.2 佈局添加_示例(列)

```
</head>
<body>

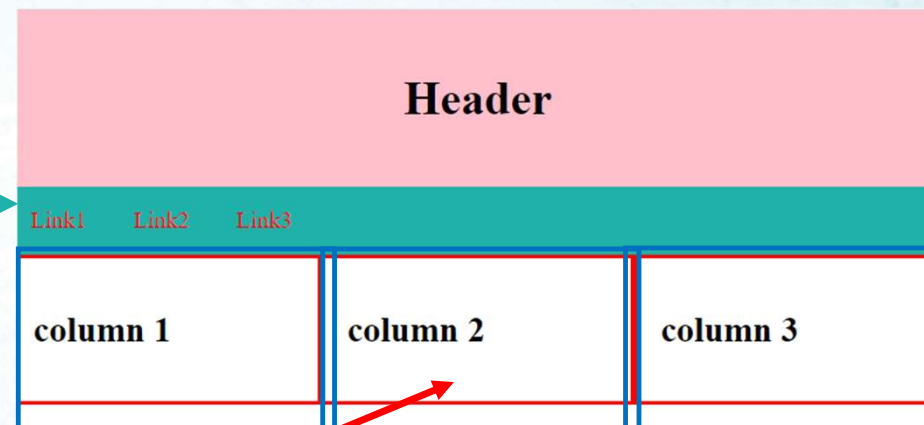
  <div class="header">
    <h1>Header</h1>
  </div>

  <div class="topnav">
    <a href="#">Link1</a>
    <a href="#">Link2</a>
    <a href="#">Link3</a>
  </div>
```

```
    <div class="row">
      <div class="column">
        <h2>column 1</h2>
      </div>

      <div class="column">
        <h2>column 2</h2>
      </div>

      <div class="column">
        <h2>column 3</h2>
      </div>
    </div>
```



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8.2 佈局添加_示例(列)

請注意，"框的大小"是必要的，否則，它將不會顯示結果。

```
<style>
* {
  box-sizing: border-box;
}
body {
  margin: 0;
}
.header {
  background-color: pink;
  padding: 20px;
  text-align: center;
}
```

```
.topnav {
  overflow: hidden;
  background-color: lightseagreen;
}
.topnav a {
  float: left;
  color: red;
  text-align: center;
  padding: 14px 16px;
  text-decoration: none;
}
.topnav a:hover {
  background-color: orange;
  color: black;
}
```

Link1 Link2 Link3		
column 1	column 2	column 3

Link1 Link2 Link3		
column 1	column 2	
column 3		

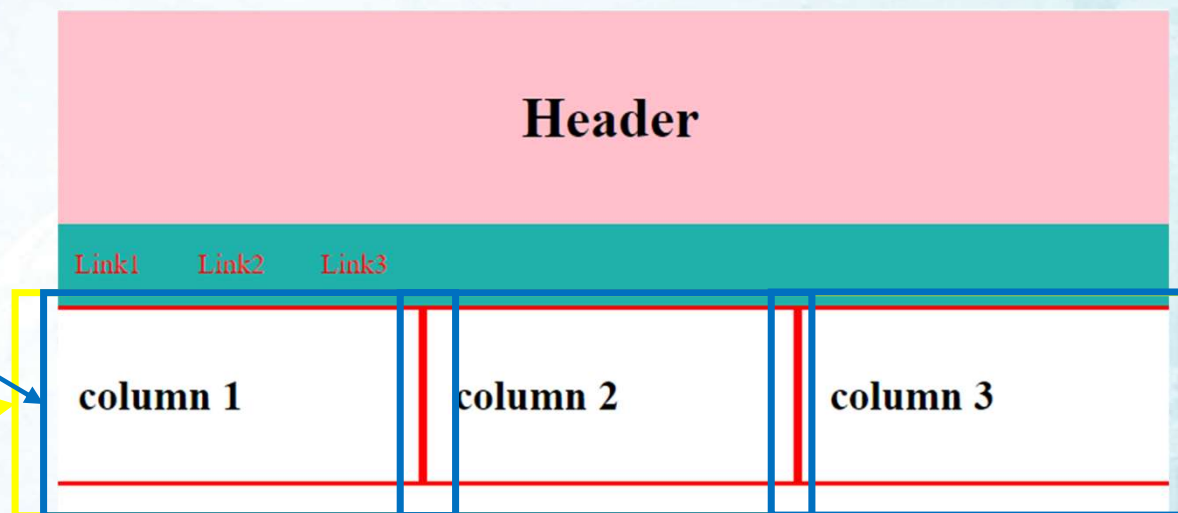
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8.2 佈局添加_示例(列)

要創建如下所示的列，".column "和".row "很重要。

```
.topnav a:hover {  
  background-color: orange;  
  color: black;  
}  
  
.column {  
  border: red solid 3px;  
  float: left;  
  width: 33.33%;  
  padding: 15px;  
}  
  
/* Clear floats after the columns */  
.row:after {  
  content: "";  
  display: table;  
  clear: both;  
}
```

</style>



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3. 疊層樣式表(CSS3)

8.3 屬性選擇器

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8.3 屬性選擇器

- 我們可以使用class/id來為元素指定一個特定的樣式
- 在本章中，它將有一個更詳細的解釋。

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8.3 屬性選擇器 示例(原文)

```
<style>
  body{
    background-color: lightcyan;
  }

  strong{
    color: red;
    font-size: 15px;
  }

  .section1{
    background-color: orange;
  }
</style>
```

This is a section **without any class**

This is section 1 =

```
</head>
<body>

  <h1> This is a section <strong> without any class
  </strong></h1>
  <h1 class="section1"> This is section 1 <strong
  style="font-size:20px;"> = </strong></h1>

</body>
```

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8.3 屬性選擇器 示例(attribute="value")

```
<style>

  body{
    background-color: lightcyan;
  }

  strong{
    color: red;
    font-size: 15px;
  }

  h1[ class="section 1" ]{
    background-color: orange;
  }

</style>
```

This is a section **without any class**

This is section 1

This is section 2

This is section 3

```
<body>

  <h1> This is a section <strong> without any class
  </strong></h1>

  <h1 class="section 1"> This is section 1 </h1>
  <h1 class="section 2"> This is section 2 </h1>
  <h1 class="section 3"> This is section 3 </h1>

</body>
```

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8.3 屬性選擇器 示例(attribute ~="value")

```
h1[ class~="section" ]{  
    background-color: orange;  
}
```

This is section 1

This is section 2

This is section 3

```
<body>  
    <h1> This is a section <strong> without any class  
    </strong></h1>  
    <h1 class="section 1"> This is section 1 </h1>  
    <h1 class="section 2"> This is section 2 </h1>  
    <h1 class="section 3"> This is section 3 </h1>  
</body>
```

注意：為屬性選擇器定義一個明確的值是很重要的
selector:"section" 而不是 "section"

8. CSS高階III

8.3 屬性選擇器 示例(attribute |= "value")

```
h1[ class |= "section" ]{  
    background-color: orange;  
}
```

This is section 1

This is section 2

This is section 3

This is section 4

當類別名稱包含完全相同的「值」時，它就能正常運作

```
<body>  
    <h1> This is a section <strong> without any class  
    </strong></h1>  
    <h1 class="section-1"> This is section 1 </h1>  
    <h1 class="section"> This is section 2 </h1>  
    <h1 class="section "> This is section 3 </h1>  
    <h1 class="section 4"> This is section 4 </h1>  
</body>
```

8. CSS高階III

8.3 屬性選擇器 示例(屬性^="值")

```
h1[ class^="section" ]{  
    background-color: orange;  
}
```

當類別名稱以"值"開始時，它是有效的

This is a section **without any class**

This is section 1

This is section 2

This is section 3

This is section 4

```
<body>  
    <h1> This is a section <strong> without any class  
    </strong></h1>  
    <h1 class="1 section"> This is section 1 </h1>  
    <h1 class="2 section 2"> This is section 2 </h1>  
    <h1 class="section 3"> This is section 3 </h1>  
    <h1 class="section 4"> This is section 4 </h1>  
</body>  
</html>
```

8. CSS高階III

8.3 屬性選擇器 示例(屬性\$="value")

```
h1[class $="section" ]{  
    background-color: orange;  
}
```

請注意，如果它以空格結尾，則無法正常工作

This is a section without any class

This is section 1

This is section 2

This is section 3

This is section 4

```
<h1> This is a section <strong> without any class  
</strong></h1>
```

```
<h1 class="1 section"> This is section 1 </h1>
```

```
<h1 class="2 section "> This is section 2 </h1>
```

```
<h1 class="section 3"> This is section 3 </h1>
```

```
<h1 class="section 4"> This is section 4 </h1>
```

```
</body>
```

```
</html>
```


8. CSS高階III

8.3 屬性選擇器 示例(屬性*="值")

```
h1[class *= "sec"] {  
    background-color: orange;  
}
```

只要類別名稱/ID包含該值，就會顯示效果

This is a section **without any class**

This is section 1

This is section 2

This is section 3

This is section 4

<body>

<h1> This is a section without any class
</h1>

<h1 class="section"> This is section 1 </h1>

<h1 class="section "> This is section 2 </h1>

<h1 class="section 3"> This is section 3 </h1>

<h1 class="section 4"> This is section 4 </h1>

</body>

8. CSS高階III

8.3 屬性選擇器 示例(屬性*="值")

```
h1[class *= "sec"] {  
    background-color: orange;  
}
```

This is a section without any class

This is section 1

This is section 2

This is section 3

This is section 4

```
<body>  
  
  <h1> This is a section <strong> without any class  
  </strong></h1>  
  
  <h1 class="section-1"> This is section 1 </h1>  
  <h1 class="123sec "> This is section 2 </h1>  
  <h1 class="se123ction"> This is section 3 </h1>  
  <h1 class="section 4"> This is section 4 </h1>  
  
</body>
```

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8.3 屬性選擇器(總結)

程式碼	意義
屬性= "值"	類別/id名稱必須完全相同 (value)
屬性~= "值"	類別/id名稱包含值(不帶連字符-) (abc值)
屬性 = "值"	類別/id名稱包含的值 (值-a)
屬性^= "值"	類別/id名稱以值開始 (值 abc/值abc/值-abc)
屬性\$= "值"	類別/id名稱以值結尾 (abc 值/abc值/abc-值)
屬性*= "值"	只要類別/id名稱包含值 (123值234)

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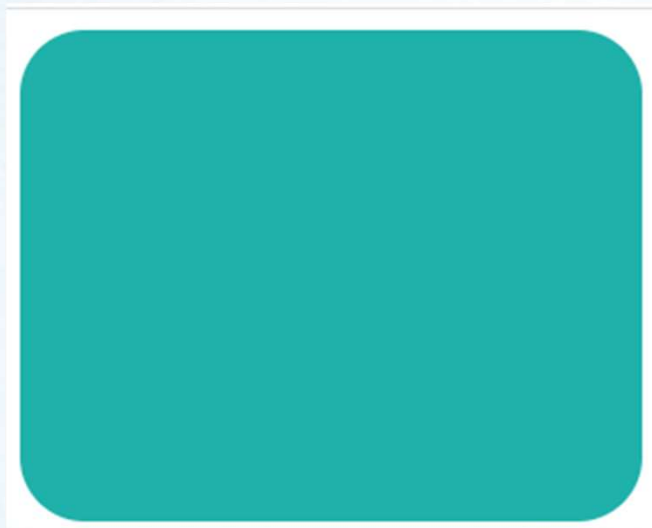
8.4 圖像設計

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8.4 圖像設計

我們可以指定一個圓角的DIV/圖片，這可以使你的網站看起來更好。



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8.4 圖像設計示例

```
.rcorners1 {  
  border-radius: 25px;  
  background: lightseagreen;  
  padding: 20px;  
  width: 200px;  
  height: 150px;  
  margin-right: 20px;  
}
```

This is section 1

```
.rcorners2 {  
  border-radius: 50px;  
  background: blue;  
  padding: 20px;  
  width: 200px;  
  height: 150px;  
}
```

This is section 2

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8.4 圖像設計示例

你也可以為"border-radius "設定不同的值，以看到不同的效果。

```
.rcorners1 {  
  border-radius: 25px 15px 30px 0px;  
  background: lightseagreen;  
  padding: 20px;  
  width: 200px;  
  height: 150px;  
  margin-right: 20px;  
}
```

This is section 1

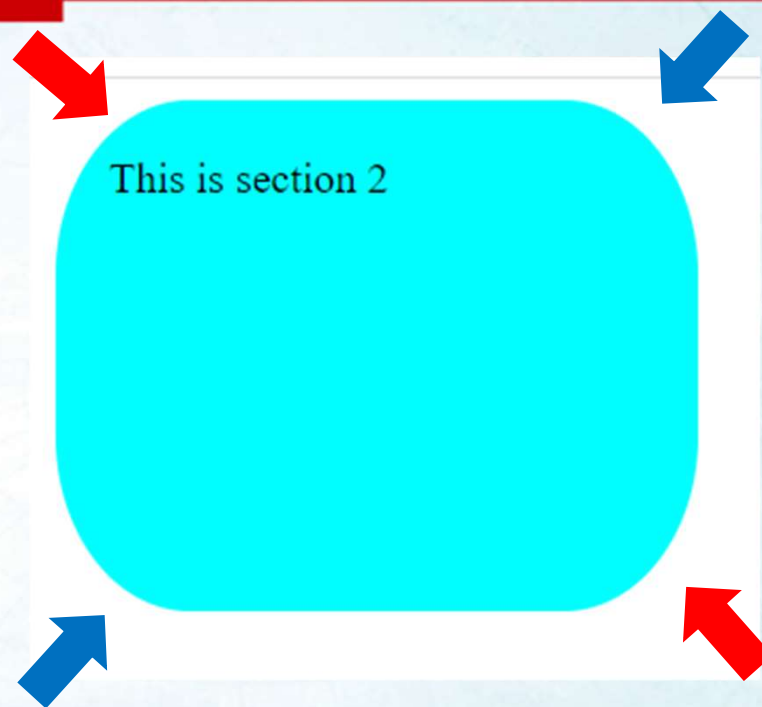
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8.4 圖片設計示例(border-radius)

```
.rcorners2 {  
  border-radius: 50px/65px;  
  background: cyan;  
  padding: 20px;  
  width: 200px;  
  height: 150px;  
}
```



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8. CSS高階III

測驗時間第1題

1. ::before{ Counter- increment: 123;

Counter: "___a___" counter(123); }

請填寫空白處以顯示以下結果。

回答：

a=Subject

Subject 1: Chinese

Subject 2: English

Subject 3: Math

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以作任何形式之翻印。

8. CSS高階III

測驗時間第2題

2. 以下哪種情況將內容分為4欄？

```
.column{ a. width:15%  
         b. width:25%  
         c. width:30%  
         d. width:40% }
```

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測驗時間第3題

3. 屬性選擇器有多少種？

- a. 4
- b. 5
- c. 6
- d. 7

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測驗時間第4題

4. 以下哪一個類的名字可以用？

a. A [class ~ = "abc"]

b. A [class | = "abc"]

c. A [class * = "abc"]

d. A [class \$ = "abc"]

1. Abc-123

2. 123abc

3. 123ABC

4. 12abc3

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8. CSS高階III

測驗時間第5題

5. 以下哪項是製作圖像圓角的正確程式碼？

- a. `Border radis:15px;`
- b. `Border radius:15px;`
- c. `Border-radius:15px;`
- d. `Border-corner:15px;`

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8. CSS高階III

參考資料

有一些有用的網站供參考:

1. https://www.w3schools.com/css/css_counters.asp
2. https://www.w3schools.com/css/css_attribute_selectors.asp
3. https://www.w3schools.com/css/css_website_layout.asp
4. https://www.w3schools.com/css/css3_borders.asp

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